

Reddy Balaji Madha

SOFTWARE ENGINEER • SYSTEMS & STORAGE

805-732-2532 | reddybalajimadha@gmail.com | LinkedIn: linkedin.com/in/reddy-balaji-madha |

GitHub: github.com/reddybalajimadha

Summary

- Software Engineer and System Administrator with enterprise SAN/NAS experience (NetApp ONTAP, Dell EMC) and hands-on EMS interoperability work (FHIR/HL7/NEMESIS). Builds secure, multi-tenant services with Elasticsearch, AWS S3, and containerized deployments (Docker).

Experience

Atian.ai (HealthyVibes.ai)

Software Developer

Remote

Dec 2024 - Present

EMS interoperability (NEMESIS / FHIR)

- Built a NEMESIS FHIR API microservice (Node.js/Express) for EMS data exchange, including EMR-to-NEMESIS mapping, validation, export logging, and error handling.
- Developed SFTP and HTTP API exporters; modeled core NEMESIS entities in Sequelize (Incident, Patient, Vitals, Medications, Procedures, Billing).

CAD/911 incident integration and multi-tenant search

- Implemented CAD webhook ingestion and multi-tenant incident management in Elasticsearch with tenant-based index resolution and CAD incident CRUD APIs.
- Built patient resolution workflows (match/link/placeholder) and integrated CAD data into pre-hospital reports to improve correlation and completeness.

Clinical reporting, PDF merge, and edge deployments

- Implemented PDF merge for ECG/CHART/pre-hospital reports with AWS S3 integration, validation, configurable merge order, and failure-safe logging.
- Delivered containerized edge deployments (MediaMTX + AI services) and an RTSP-to-transcription pipeline using FFmpeg decoding, buffering, and retries for unstable networks.

Tata Consultancy Services (TCS)

System Administrator

Bangalore, India

Feb 2020 - Aug 2022

Enterprise storage operations

- Administered SAN/NAS storage (NetApp ONTAP, Dell EMC) including provisioning, monitoring, troubleshooting, and capacity planning for production systems.
- Led an annual home-directory migration project coordinating ~100 requests to move data to the nearest site, minimizing downtime and improving user performance.
- Supported Cisco MDS and VMware storage connectivity; troubleshoot network/storage issues using Wireshark and runbooks.

Education

Jessup University

M.Sc. in Computer Science

2025 - 2027

California Lutheran University

M.S. in Information Technology (GPA: 3.74/4.0)

2022 - 2024

Skills

Languages	Java, Python, SQL, Bash, TypeScript
Backend	FastAPI, Node.js/Express, REST APIs, Webhooks, Pydantic/jsonschema
Datastores	Elasticsearch, Cassandra, SQLite, MySQL/PostgreSQL (basics), Sequelize ORM
Cloud/DevOps	AWS (S3, EC2, EFS), Docker, Docker Compose, GitHub Actions, Kubernetes configs
Storage/Net	NetApp ONTAP, Dell EMC, SAN/NAS, Cisco MDS, VMware concepts, Wireshark